25900686 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGY200 - ETHOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Cerebrum is a part of……. [K] / [CO2]

   a). Forebrain b). Midbrain

   c). Hindbrain d). limbic system

2. Almond shaped structure in brain is…… [K] / [CO2]

   a). Hypothalamus b). Amygdala

   c). Hippocampus d). Cerebellum

3. Which reproduction strategy would be favored in a resource rich environment? [An] / [CO3]

   a). k-selection strategy b). r-selection strategy

   c). None of the above d). Both a and b

4. Trophallaxis in ants is: [K] / [CO4]

a). Food storage in chambers b). Exchange of food mouth-to-mouth

c). Waggle dance communication d). None of the above

5. Which hormone is known as the 'stress hormone'? [U] / [CO3]

a). Oxytocin b). Cortisol

c). Insulin d). Glucagon

6. Which motor area is responsible for programming sequences of movement? [K] / [CO2]

a). Supplementary motor area b). Frontal eye field

c). Wernicke’s area d). Angular gyrus

7. Select the primary cost associated with large groups. [K] / [CO4]

a). Decreased social interaction b). Higher competition for food

c). Reduced predator detection d). Increased reproductive success

8. Male-male competition is usually seen in which type of sexual selection? [U] / [CO3]

a). Intersexual selection b). Only a

c). Intrasexual selection d). None of the above

9. Illustrate the adaptive value of territorial marking. [K] / [CO4]

a). Increases chances of injury b). Prevents reproduction

c). Reduces direct conflict and saves energy d). Eliminates group survival

10. Ethology helps in animal reintroduction programs by: [K] / [CO1]

a). Training survival behaviors b). Selecting individuals fit for the wild

c). Both A and B d). B only

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11.	Compare the role of estrogen and testosterone in parental behaviour.	[An] / [CO3]
12.	Differentiate between explicit and implicit memory.	[U] / [CO2]
13.	Explain the two major mechanisms behind coloration in animals	[U] / [CO3]
14.	Give any two examples where reciprocal altruism is observed.	[U] / [CO4]
15.	Discuss learning centres in brain.	[U] / [CO2]
16.	How did I.P.Pavlov's work show the role of conditioning in shaping behavior?	[U] / [CO1]
17.	What motor deficits occur with damage to the premotor cortex?	[K] / [CO2]
18.	How does group foraging differ from solitary foraging?	[U] / [CO4]
19.	A male bird continues to sing despite the increased risk of attracting predators. Analyze this behavior in terms of the trade-off between survival and reproductive success. What does this suggest about the evolutionary pressures shaping communication strategies in animals?	[An] / [CO5]
20.	Compare aestivation and hibernation.	[An] / [CO3]
21.	Interpret EEG in REM vs NREM sleep.	[U] / [CO3]
22.	Critically analyze limitation of Lorenz's psycho-hydraulic model.	[U] / [CO2]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23.	Analyze how male scorpions use behavioral adaptations to safely engage with potentially aggressive females.	[A] / [CO3]
24.	Explain different grouping patterns in primates	[K] / [CO4]
25.	Explain latent learning in the light of Tolman's experiment.	[U] / [CO2]

26.	Compare and contrast Batesian and Müllerian mimicry, highlighting their differences and similarities.	[An] / [CO3]
27.	Explain in detail on Taste aversion with examples	[K] / [CO2]
28.	Compare different types of parenting in animals.	[An] / [CO3]
29.	Discuss various biological aspects of pheromons.	[U] / [CO5]
30.	Analyse tidal rhythms in coastal organisms and connect them to survival strategies.	[An] / [CO3]

[5x6 = 30]